UC San Diego

Trilinear estimates for quadratic dispersive equations

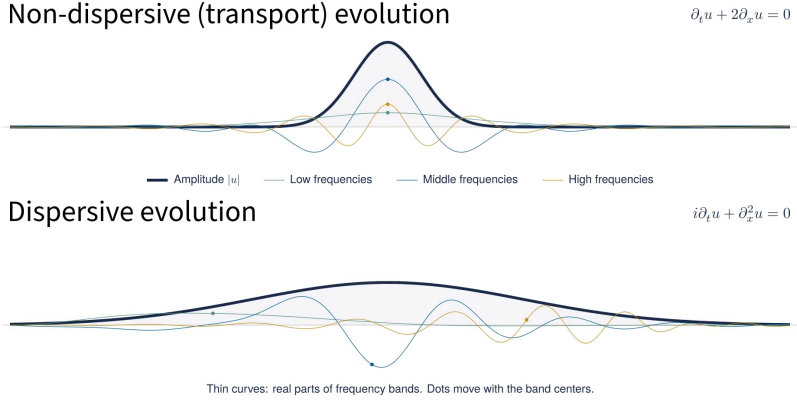
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includes work with Ioan Bejenaru and Tiklung Chan
University of California San Diego



UC Berkeley – HADES
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Introduction

What is dispersion?



Different frequencies move at different speeds, and the wave disperses!

Quadratic semilinear equations

Schematically, a quadratic and dispersive equation has the form

$$(i\partial_t + L(D))u = N(u),$$
$$u(0) = u_0$$

Examples: Schrödinger (Δ), Klein–Gordon ($\langle D \rangle$), wave ($|D|$).

Physics: optics, quantum mechanics, gravitation, water waves…

Motivating question

For which initial data do we get global-in-time solutions? How do these solutions behave as $t \to \infty$?

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Under what conditions do small initial data u_0 give rise to global-in-time solutions that asymptotically approach free waves?

$$\|u(t) - e^{itL(D)}u_*\|_{H^s_x} \longrightarrow 0 \quad \text{as} \quad t \to \infty.$$

where $u_{\text{free}}(t) = e^{itL(D)}u_*$ solves $(i\partial_t + L(D))u_{\text{free}} = 0$ with $u_{\text{free}}(0) = u_*$.

Why should one expect scattering?

$$(i\partial_t - \Delta)u = \underbrace{N(u)}_{\text{nonlinear forcing}}$$

Small accumulated forcing $\longrightarrow$ small correction to free evolution.

Linear dispersion

$$(i\partial_t - \Delta)u_{\text{free}} = 0$$

$$\|u_{\text{free}}(t)\|_{L^\infty_x} \lesssim (1+t)^{-n/2}.$$

Decay weakens quadratic forcing.

Possible ODE buildup

$$\partial_t v = v^2$$

$$v(t) = \frac{1}{T-t}.$$

Blows up as $t \longrightarrow T^-$.

> **Can dispersion win?** For small data, hope the late-time forcing becomes negligible and the solution approaches a free wave.

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Literature

Schrödinger equations

Germain–Masmoudi–Shatah (2009)
Bejenaru–Ionescu–Kenig–Tataru (2011)
Herr–Tataru–Tzvetkov (2011)
Ifrim–Tataru (2015)
Ifrim–Tataru (2023)
Ifrim–Tataru (2024)

Klein–Gordon equations

Shatah (1985)
Bejenaru–Herr (2017)
Chae–Oh (2017)
Ionescu–Pausader (2019)
Bejenaru–Borges (2026)
Borges–Chan (2026)

Wave equations and GR

Christodoulou–Klainerman (1993)
Tao (2001)
Tataru (2001)
Rodnianski–Tao (2004)
Lindblad–Rodnianski (2010)
Krieger–Sterbenz–Tataru (2015)
Krieger–Tataru (2017)
Ionescu–Pausader (2022)

Water waves

Wu (2011)
Germain–Masmoudi–Shatah (2012)
Alazard–Delort (2015)
Ionescu–Pusateri (2015)
Ifrim–Tataru (2016)
Deng–Ionescu–Pausader–Pusateri (2017)
Ionescu–Pusateri (2018)

Main result

Theorem (Bejenaru–B.–Chan)

The Schrödinger equation

$$(i\partial_t - \Delta)u = c_1 u^2 + c_2 \bar{u}^2, \qquad c_1, c_2 \in \mathbb{C}, \tag{2NLS}$$

with small initial data $u_0 \in \dot{H}^{-\frac{1}{2}}(\mathbb{R}^3)$ in the scaling-critical space is globally well-posed, scatters to free solutions, and has Lipschitz dependence on its initial data. The solution is also unique in the resolution space $\dot{X}^{-\frac{1}{2}} \subset C_t \dot{H}_x^{-\frac{1}{2}}$.

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> **Goal:** Explore the proof of scattering for $(i\partial_t - \Delta)u = u^2$

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(v) Everything coming together

Contraction argument

$$(i\partial_t + L(D))u = F \qquad (F = N(u))$$

Goal: Find a complete space $X \subset C_t H^s_x$, and a forcing space Y such that

- If $\|u(0)\|_{H^s_x}$ is small, then $e^{itL(D)}u_0 \in X$
- Linear estimate: $\|u\|_X \lesssim \|u_0\|_{H^s_x} + \|F\|_Y$
- Nonlinear estimate: $\|N(u) - N(v)\|_Y \lesssim (\|u\|_X + \|v\|_X)^{p-1} \|u - v\|_X$

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For small initial data, it is enough to establish

$$\|u\|_X \lesssim \|u_0\|_{H^s_x} + \|N(u)\|_Y, \quad \text{and} \quad \|N(u)\|_Y \lesssim \|u\|_X^p \tag{1}$$

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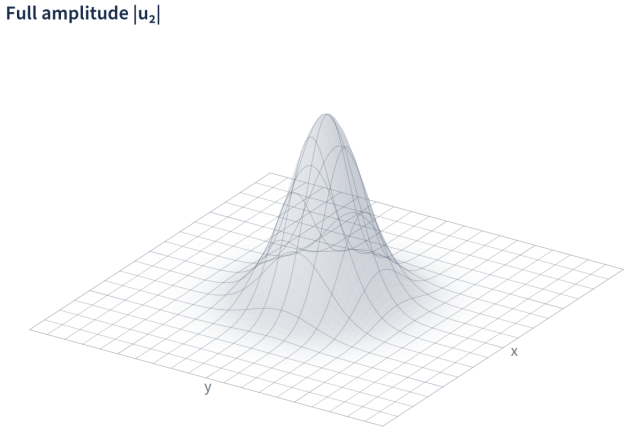
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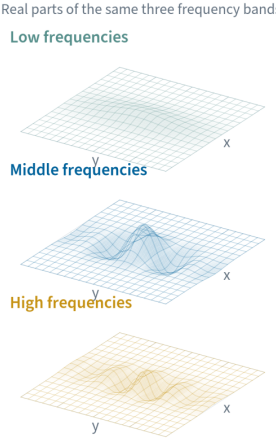
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In fact, if X is a *reasonable* space, then (1) establishes both global well-posedness and scattering.

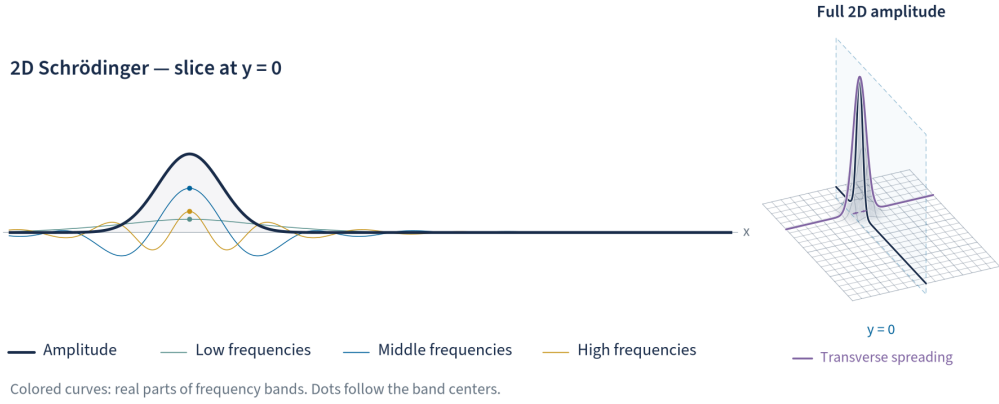
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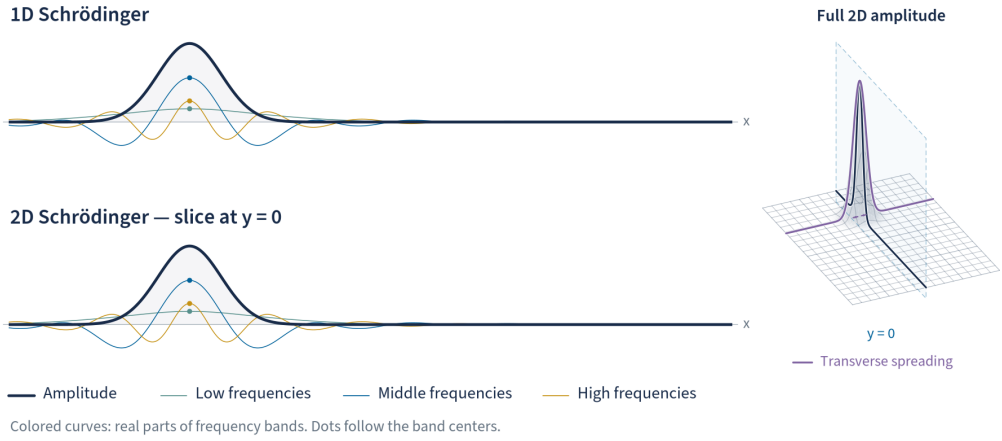
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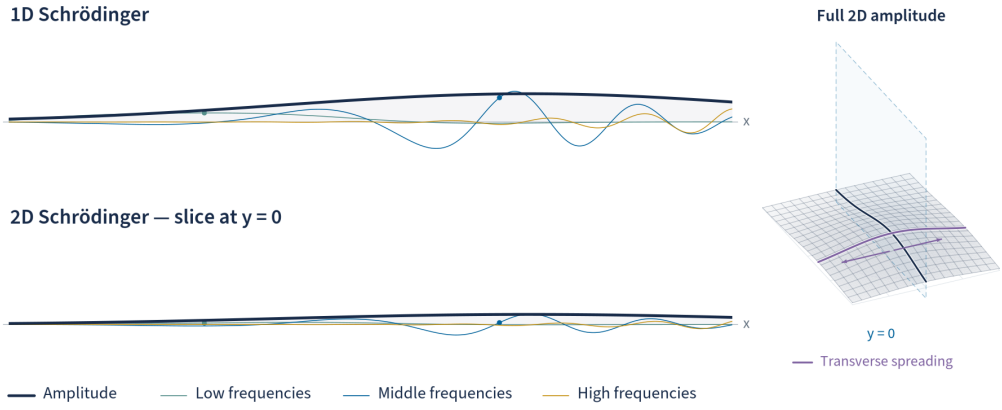
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Linear Schrödinger: Strichartz estimates

Free evolution: $\|e^{-it\Delta}u_0\|_{L_t^q L_x^r} \lesssim \|u_0\|_{L_x^2}, \qquad \frac{2}{q}+\frac{n}{r}=\frac{n}{2}, \quad 2 \le q, r \le \infty$

With forcing: if $(i\partial_t - \Delta)u = F$, then

$$\|u\|_{L_{t,x}^p} \lesssim \|u_0\|_{L_x^2} + \|F\|_{L_{t,x}^{p'}}, \qquad p = \frac{2(n+2)}{n}.$$

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Toy model: quadratic Schrödinger in dimension 4

$$(i\partial_t - \Delta)u = u^2, \quad u : \mathbb{R}_t \times \mathbb{R}^n_x \to \mathbb{C}.$$

Goal: Construct spaces X and Y such that $\|u^2\|_Y \lesssim \|u\|_X^2$.
By duality, it is enough to establish

$$|\langle u^2, w\rangle| \lesssim \|u\|_X^2 \|w\|_{Y^*}, \qquad w \in Y^*.$$

Strichartz estimates: $\|u\|_{L^p_{t,x}} \lesssim \|u_0\|_{L^2_x} + \|u^2\|_{L^{p'}_{t,x}}$, $p = \frac{2(2+n)}{n}$

$n = 4$	$p = 3$	$L^3 \cdot L^3 \cdot L^3 \subset L^1_{t,x}$	✓
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Toy model: quadratic Schrödinger in dimension 4

$$(i\partial_t - \Delta)u = u^2, \quad u : \mathbb{R}_t \times \mathbb{R}^n_x \to \mathbb{C}.$$

Goal: Construct spaces X and Y such that $\|u^2\|_Y \lesssim \|u\|_X^2$.
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First nonlinear iteration

> **Guiding principle:** nonlinear interactions carry more information than the linear theory provides

Consider the equation

$$(i\partial_t - \Delta)u = vw$$

where $v = e^{-it\Delta}g$, $w = e^{-it\Delta}h$ are free solutions localized at frequency 1.

In (spacetime) Fourier variables

$$\widehat{u}(\tau, \xi) = -\frac{\widehat{(vw)}}{\tau - |\xi|^2}$$

This is well-defined when $\widehat{v} * \widehat{w}$ is supported *away* from the surface $\tau = |\xi|^2$. Such interactions are called *nonresonant*.

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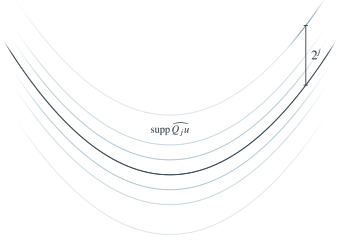
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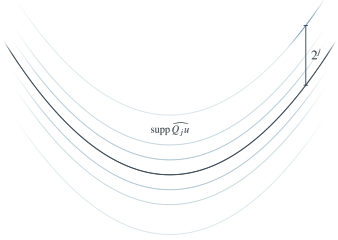
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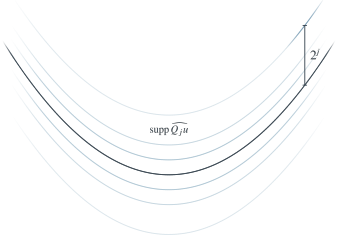
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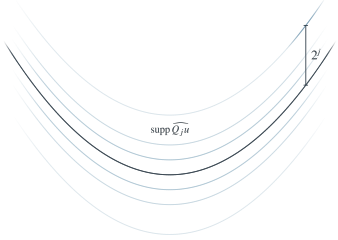
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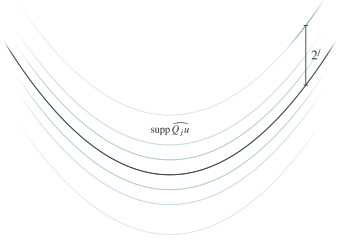
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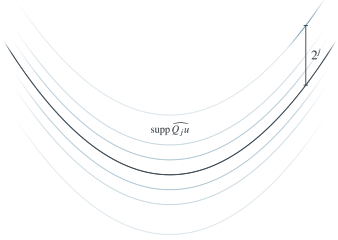
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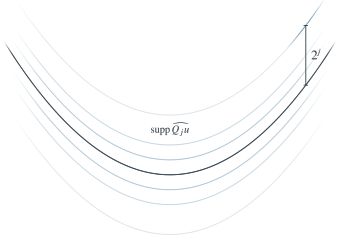
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Nonresonant Dirac–Klein–Gordon in 2D

Theorem (Bejenaru–B. ’26)

Let $M, m > 0$ with $2M > m > 0$. For small initial data

$$(\psi_0, \phi_0, \phi_1) \in H_x^{\frac{1}{2}+\epsilon}(\mathbb{R}^2) \times H_x^{1+\epsilon}(\mathbb{R}^2) \times H_x^{\epsilon}(\mathbb{R}^2),$$

the *nonresonant* system

$$(-i\gamma^\mu \partial_\mu + M)\psi = \phi\psi$$
$$(\Box + m^2)\phi = \langle \psi, \gamma^0 \psi \rangle$$

is globally well-posed and scatters.

Recap (trilinear formulation and resonance decomposition)

Our goal is to find spaces X and Y such that $|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}, \qquad w \in Y^*.$

- In high dimensions $n \geq 4$, Strichartz estimate on $L^3_{t,x}$ suffices.
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separately.
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Since u_{res} is close to a free solution, we first consider the case where u, v, w are free waves.

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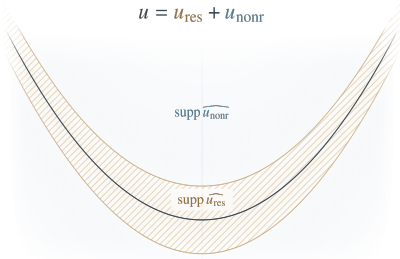
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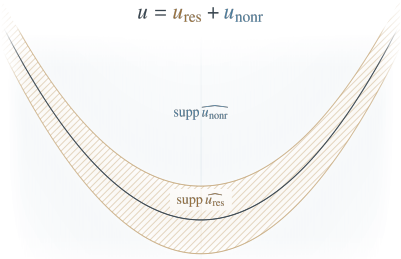


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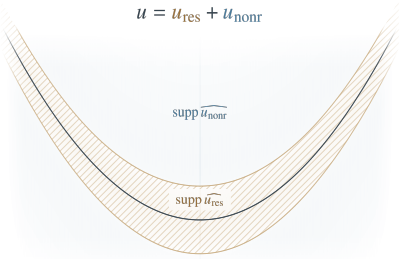


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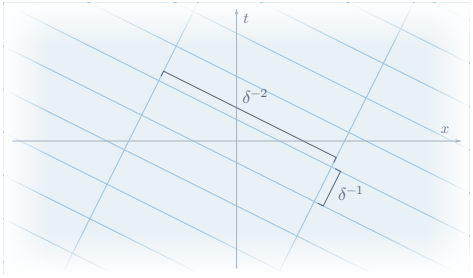
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- Since u_{nonr} is not a free solution, we have access to extra integrability.

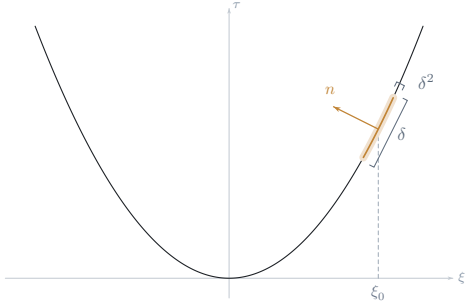


Since u_{res} is close to a free solution, we first consider the case where u, v, w are free waves.

Wave packets

Let $u = e^{-it\Delta} f$ be a free wave. In Fourier space, $\widehat{u}$ is supported on the paraboloid $\tau = |\xi|^2$:

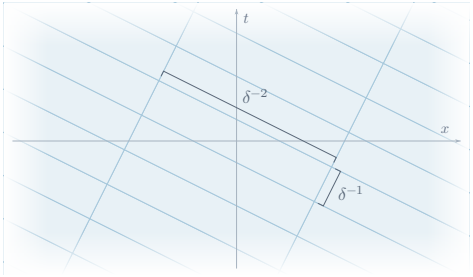


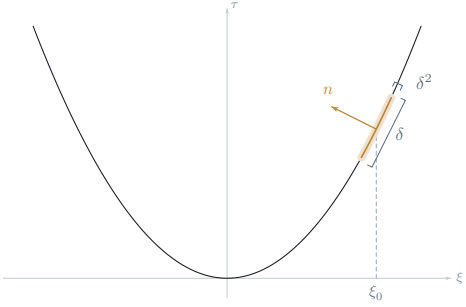


u is approximately constant on tubes of dimensions $(\delta^{-1})^3 \times \delta^{-2}$ oriented along N.

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Two-wave interactions

Question: In 3D, Hölder + Strichartz give

$$\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^{5/3}_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}.$$

Can we do better?

If N_1 and N_2 are *transversal*, $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^p_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}, \quad p > \frac{3}{2}$

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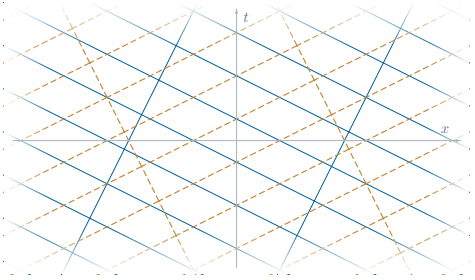
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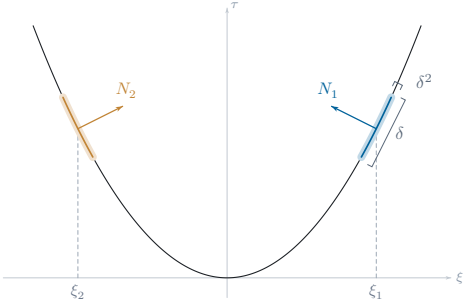
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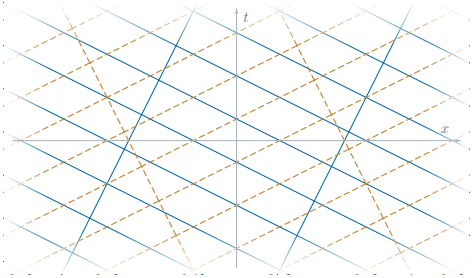
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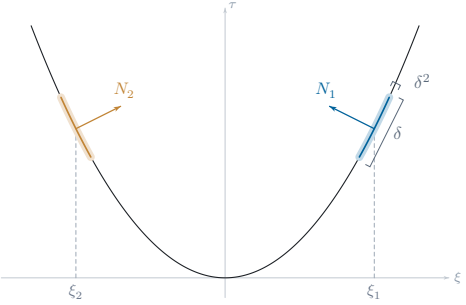
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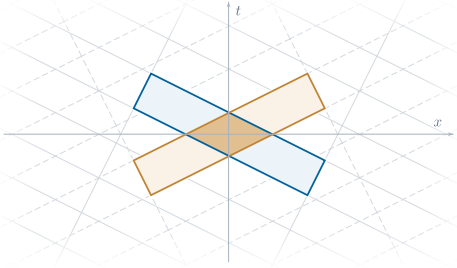


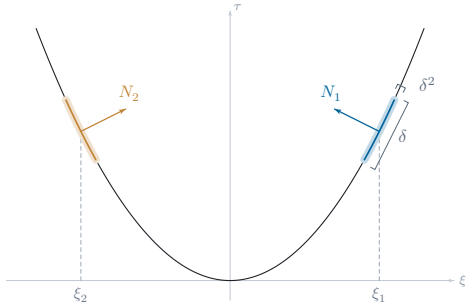


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Why is transversality important?

Heuristic: “transversal waves interact in smaller regions and for less time”



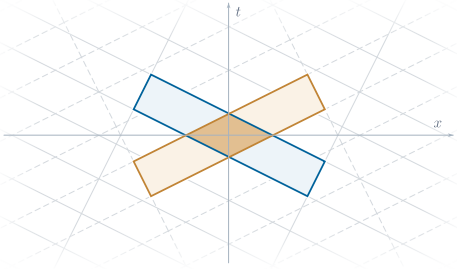


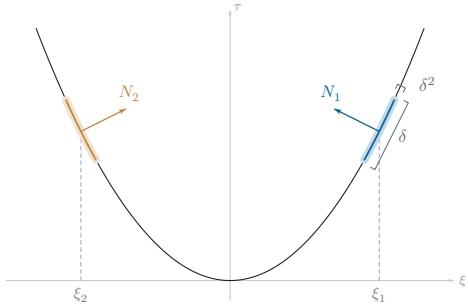
If the normals satisfy $|N_1 \wedge N_2| \geq \nu > 0$, then

$$\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^p_{t,x}} \lesssim \nu^{\frac{1}{p}-1} \|f\|_{L^2_x} \|g\|_{L^2_x}, \qquad \frac{3}{2} < p \leq 2.$$

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Bilinear estimates and the remaining resonances

For three free Schrödinger waves in dimension 3, the direct Hölder route is

$$\left|\int uv\bar{w}\ dx\,dt\right| \leq \|uv\|_{L^q_{t,x}} \|w\|_{L^{10/3}_{t,x}}, \qquad \frac{1}{q}+\frac{3}{10}=1.$$

We need a bilinear estimate with $q=\frac{10}{7}$, however bilinear restriction requires $q>\frac{3}{2}>\frac{10}{7}$

> **Takeaway:** the bilinear restriction range is not enough to control resonant quadratic interactions!

Bilinear suffices in the radial class: In that case, the correct Strichartz space is $L^{14/5}_{t,x}$, and so

$$\left|\int uv\bar{w}\ dx\,dt\right| \leq \|uv\|_{L^q_{t,x}} \|w\|_{L^{14/5}_{t,x}},$$

and the dual space $q=\frac{14}{9}>\frac{3}{2}$ belongs to the bilinear restriction range!

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The 3D Klein–Gordon–Schrödinger system

Theorem (B.–Chan, ‘26)

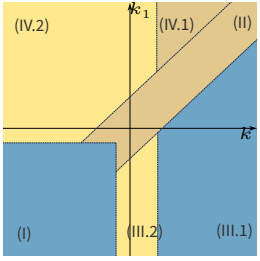
$$(i\partial_t - \Delta)u = un, \qquad (\Box + 1)n = \pm|u|^2.$$

For any $0 < \epsilon \ll 1$, sufficiently small **radial** data

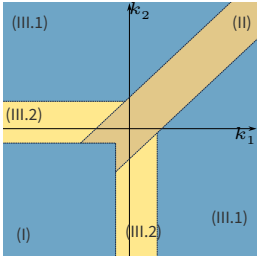
$$(u_0, n_0, n_1) \in L^2_x(\mathbb{R}^3) \times H^{-\frac{1}{2}+\epsilon}_x(\mathbb{R}^3) \times H^{-\frac{3}{2}+\epsilon}_x(\mathbb{R}^3)$$

give a global solution which scatters to free solutions as $t \to \pm\infty$.

Schrödinger equation



Klein–Gordon equation



Color key

- Blue: nonresonant
- Yellow: bilinear restriction
- Gold: radial

Can we do better than bilinear estimates?

Hölder tells you how to estimate $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^p_{t,x}}$ using only *linear* information.

Bilinear estimates use nonlinear information (transversality) to improve integrability $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^q_{t,x}}$ with some $q < p$.

Question: Do *trilinear* interactions capture any extra information compared to bilinear?

Yes!

$$\Big\| \prod_{j=1}^{3} e^{-it\Delta} f_j \Big\|_{L^p_{t,x}} \lesssim \prod_{j=1}^{3} \|f_j\|_{L^2_x}, \quad p > \frac{14}{15}$$

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Caveat: For general surfaces, transversality does not improve the range below $p = 1$.

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Transversality gives bilinear $L^2_{t,x}$.

Surfaces on $\mathbb{R}^{1+2}$

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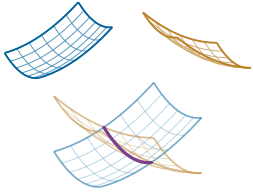
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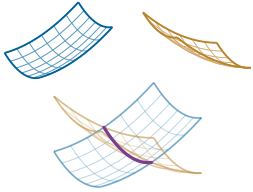
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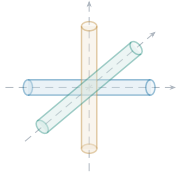
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(ii) Why do simpler approaches fail when studying (2NLS)?

(iii) Harmonic analysis: geometry of interacting waves

(iv) From harmonic analysis to PDEs

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Transference spaces

Recall that our original goal was to find spaces X and Y satisfying:

- Linear estimate: $\|u\|_X \lesssim \|u(0)\|_{L^2_x} + \|F\|_Y$,
- Multilinear estimate: $|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}$

We have discussed two types of multilinear estimates:

$$|\langle vw, u_{\text{nonr}}\rangle| \lesssim \|v\|_X \|w\|_X \|u_{\text{nonr}}\|_{L^2_{t,x}},$$

$$|\langle vw, u_{\text{res}}\rangle| \lesssim \|vw\|_{\text{Bil}} \|u_{\text{res}}\|_{\text{Bil}^*},$$

in the case where u_{res} is a free solution.

Question: How do we combine all this information into spaces X and Y?

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$$|\langle vw, u_{\text{nonr}}\rangle| \lesssim \|v\|_X \|w\|_X \|u_{\text{nonr}}\|_{L^2_{t,x}},$$

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in the case where u_{res} is a free solution.

Question: How do we combine all this information into spaces X and Y?

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Recall that our original goal was to find spaces X and Y satisfying:

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Let

$$X = L^2_t L^6_x \cap L^\infty_t L^2_x, \quad Y = L^2_t L^{6/5}_x + L^1_t L^2_x.$$

Since $Y^* = X$, the trilinear estimate reduces to

$$\Big| \int_{\mathbb{R}^{1+3}} uv\bar{w}\, dx\, dt \Big| \lesssim \|u\|_X \|v\|_X \|w\|_X$$

Pros

- Easy to define
- Easy to establish duality

Cons

- Only considers linear information
- Not enough to close the contraction when $n \leq 3$
- It does not allow for (bilinear) $L^2_{t,x}$

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$$e^{-it\Delta}f = \mathbb{1}_{[0,\infty)}(t)e^{-it\Delta}f, \qquad t \geq 0$$



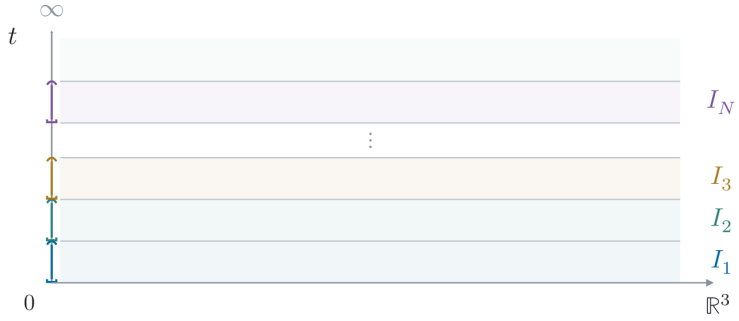
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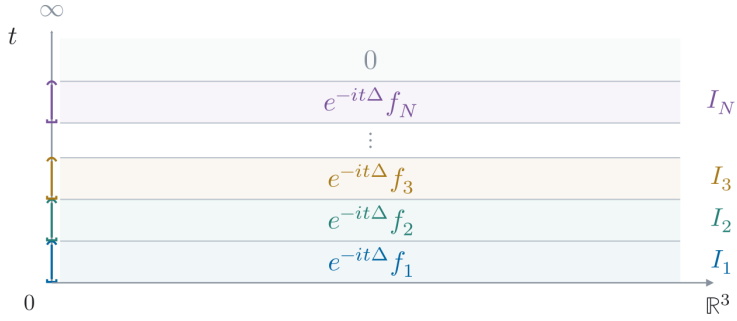
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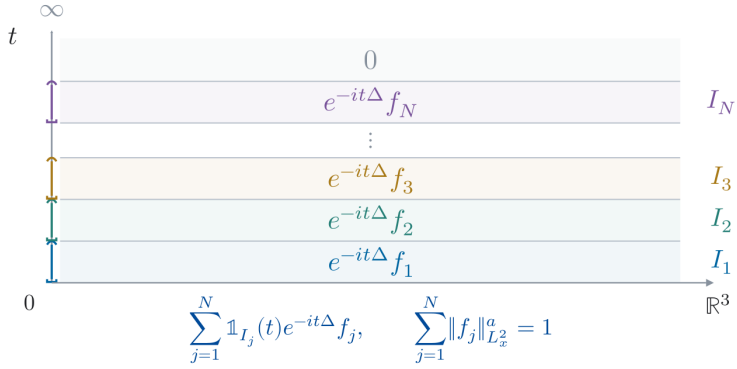
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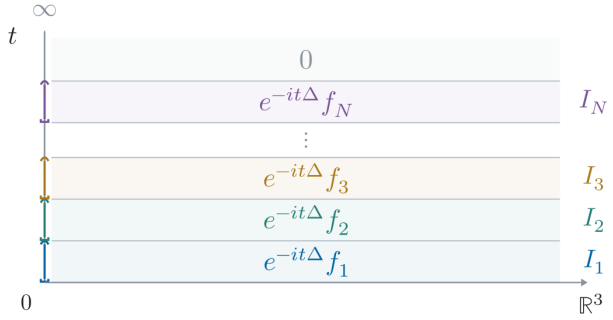
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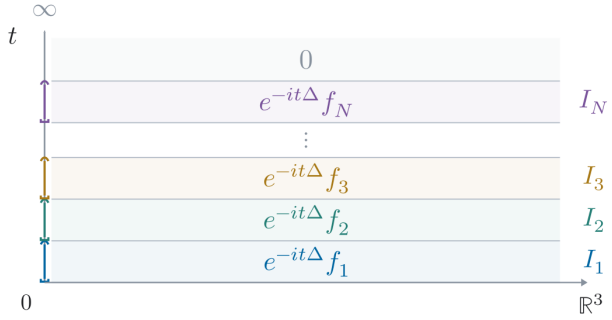
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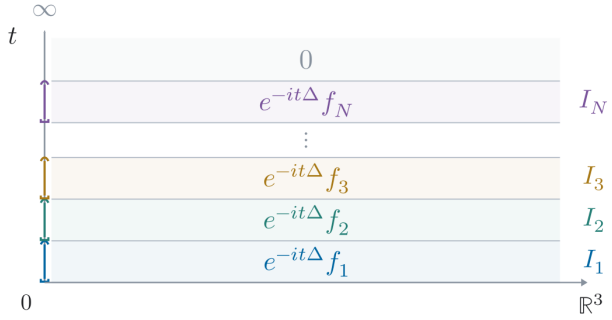
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Transference from free solutions

U^2_Δ-functions are "almost free solutions" and most estimates for free solutions also hold in this space:

- Linear: if $\|e^{-it\Delta} f\|_{L^p_t L^q_x} \lesssim \|f\|_{L^2_x}$ for all $f \in L^2_x$, then $\|u\|_{L^p_t L^q_x} \lesssim \|u\|_{U^2_\Delta}$ for all $u \in U^2_\Delta$.
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U^2_Δ is almost the perfect space... Except for duality.

$$|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}, \qquad U^2_\Delta \subset V^2_\Delta$$

Estimate we need

$$\left| \int uv\bar{w}\, dx\, dt \right| \lesssim \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta} \|w\|_{V^2_\Delta}$$

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$$\left| \int uv\bar{w}\, dx\, dt \right| \lesssim \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta} \|w\|_{V^2_\Delta}, \qquad U^2_\Delta \subset V^2_\Delta$$

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Transference from free solutions

U^2_Δ-functions are "almost free solutions" and most estimates for free solutions also hold in this space:

- Linear: if $\|e^{-it\Delta} f\|_{L^p_t L^q_x} \lesssim \|f\|_{L^2_x}$ for all $f \in L^2_x$, then $\|u\|_{L^p_t L^q_x} \lesssim \|u\|_{U^2_\Delta}$ for all $u \in U^2_\Delta$.
- Bilinear: if $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^q_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}$, then $\|uv\|_{L^q_{t,x}} \lesssim \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta}$
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Goal: Explore the proof of scattering for $(i\partial_t - \Delta)u = u^2$

(i) The building blocks: contraction theorems and high-dimensional waves

(ii) Why do simpler approaches fail when studying (2NLS)?

- Slow dispersion in 3d, Insufficient time integrability
- Bilinear range is insufficient in general

(iii) Harmonic analysis: geometry of interacting waves

- Linear wave packets
- Bilinear range is very good in the radial case
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Main result: revisited

Theorem (Bejenaru–B.–Chan)

The Schrödinger equation

$$(i\partial_t - \Delta)u = c_1 u^2 + c_2 \bar{u}^2, \qquad c_1, c_2 \in \mathbb{C}, \tag{2NLS}$$

with small initial data $u_0 \in \dot{H}^{-\frac{1}{2}}(\mathbb{R}^3)$ in the scaling-critical space is globally well-posed, scatters to free solutions, and has Lipschitz dependence on its initial data. The solution is also unique in the resolution space $\dot{X}^{-\frac{1}{2}} \subset C_t \dot{H}_x^{-\frac{1}{2}}$.

In order to prove this, we need to:

- Classify the resonances
- Check whether resonant interactions are transversal
- Split $u = u_{\text{res}} + u_{\text{nonr}}$ and establish

$$\left| \int_{\mathbb{R}^{1+3}} u_{\text{nonr}} v \bar{w} \, dx \, dt \right| \lesssim \|u\|_{V_\Delta^2} \|v\|_{V_\Delta^2} \|w\|_{V_\Delta^2}, \qquad \left| \int_{\mathbb{R}^{1+3}} u_{\text{res}} v \bar{w} \, dx \, dt \right| \lesssim \|u\|_{V_\Delta^2} \|v\|_{V_\Delta^2} \|w\|_{V_\Delta^2}$$

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Resonances for $(i\partial_t - \Delta)u = u^2$

$$\int_{\mathbb{R}^{1+3}} uv\bar{w}\,dx\,dt = \left(\hat{u} * \hat{v} * \hat{\bar{w}}\right)(0) = \int \hat{u}(\xi,\tau)\hat{v}(\eta,\tilde{\tau})\overline{\hat{w}}(\xi+\eta,\tau+\tilde{\tau})\,d\xi\,d\eta\,d\tau\,d\tilde{\tau}$$

Resonant interactions occur exactly when u, v, and w are free waves, which are (Fourier) supported on the paraboloid $\tau = \xi^2$. So resonances occur when $\xi^2 + \eta^2 = (\xi+\eta)^2$, that is, when $\xi \cdot \eta = 0$.

To check transversality, we compute $|N_\xi \wedge N_\eta \wedge N_{\xi+\eta}| \approx \min\{|\xi|, |\eta|\}$, which degenerates at low frequencies!

Resonances for $(i\partial_t - \Delta)u = u^2$

$$\int_{\mathbb{R}^{1+3}} uv\bar{w}\,dx\,dt = \left(\hat{u} * \hat{v} * \hat{\bar{w}}\right)(0) = \int \hat{u}(\xi,\tau)\hat{v}(\eta,\tilde{\tau})\overline{\hat{w}}(\xi+\eta,\tau+\tilde{\tau})\,d\xi\,d\eta\,d\tau\,d\tilde{\tau}$$

Resonant interactions occur exactly when u, v, and w are free waves, which are (Fourier) supported on the paraboloid $\tau = \xi^2$. So resonances occur when $\xi^2 + \eta^2 = (\xi+\eta)^2$, that is, when $\xi \cdot \eta = 0$.

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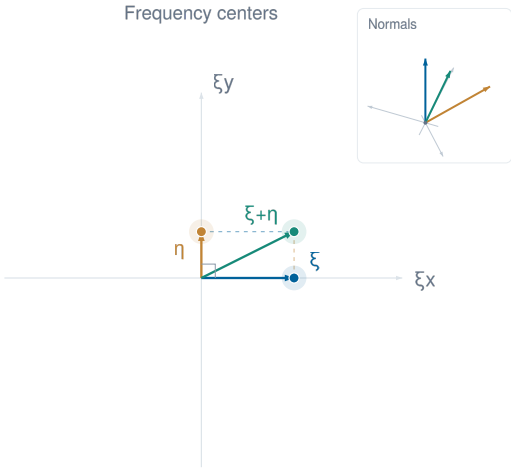
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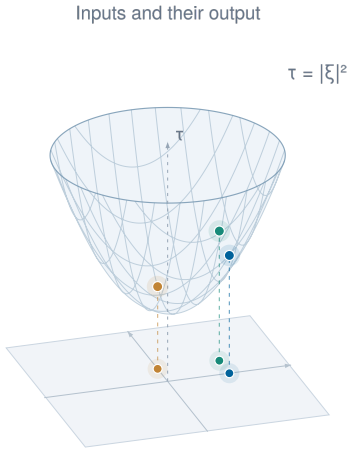
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Three-wave interactions





Nonresonant estimates

We can assume that v, w have frequency 1 and u has frequency 2^k, with $k \leq 0$.
Let $Q_j u$ localize u to the region $|\tau - \xi^2| \approx 2^j, j \in \mathbb{Z}$.

High modulation on u

High modulation on v

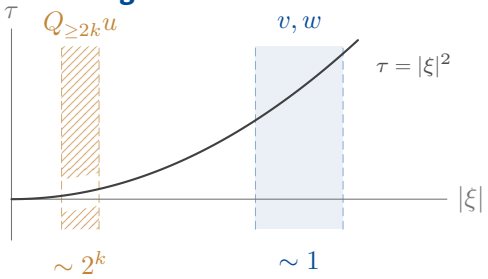
Schematic frequency–modulation slices; hatched regions have $|\tau - |\xi|^2| \gtrsim 2^{2k}$.

Claim: $\left|\int_{\mathbb{R}^{1+3}} (Q_{\geq 2k}u) v \bar{w}\, dx\, dt\right| + \left|\int_{\mathbb{R}^{1+3}} u (Q_{\geq 2k}v) \bar{w}\, dx\, dt\right| \lesssim \langle k \rangle^3 \|u\|_{V^2_\Delta} \|v\|_{V^2_\Delta} \|w\|_{V^2_\Delta}$

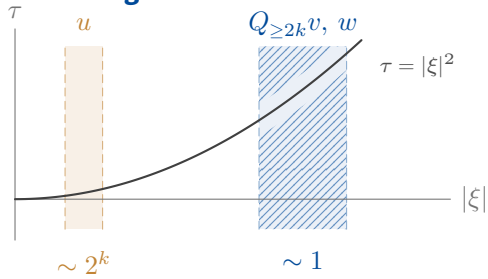
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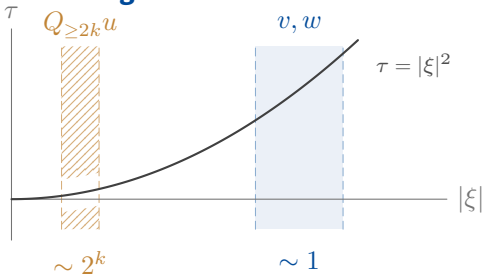
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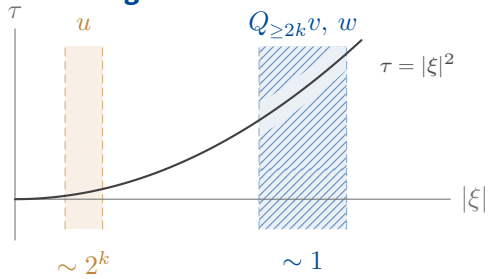
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Techniques: bilinear $L^2_{t,x}$, $\dot{X}^{0,1/2}$, Fourier cube decomposition

Resonant estimates

Goal: $\left| \int_{\mathbb{R}^{1+3}} Q_{<2^k} u \cdot Q_{<2^k} v \cdot \overline{Q_{<2^k} w}\, dx\, dt \right| \leq C_\varepsilon 2^{-\varepsilon k} \|u\|_{V^2_\Delta} \|v\|_{V^2_\Delta} \|w\|_{V^2_\Delta}$

Step 1: Establish the free wave estimate

$$\left| \int_{\mathbb{R}^{1+3}} e^{-it\Delta} f \cdot e^{-it\Delta} g \cdot \overline{e^{-it\Delta} h}\, dx\, dt \right| \leq C_\varepsilon 2^{-k\varepsilon} \|f\|_{L^2_x} \|g\|_{L^2_x} \|h\|_{L^2_x}$$

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Trilinear transference to V^2_Δ

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V^2_Δ does not have an atomic structure, and it is hard to work with directly. So instead we show that

$$\|Q_{<2k}u \cdot Q_{<2k}v \cdot \overline{Q_{<2k}w}\|_{\ell^p L^1(\mathbb{R}^{1+3})} \leq C_\varepsilon 2^{-\varepsilon k} \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta} \|w\|_{U^2_\Delta}$$
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for some $\frac{14}{15} < p < 1$ and $a > 2$. Recall that $U^2_\Delta \subset V^2_\Delta \subset U^a_\Delta$, and so V^2_Δ-transference follows from interpolation.

The fact that we can prove an estimate for $p < 1$ is crucial for the interpolation, which means that the curvature of the paraboloid plays a key role in this result.

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Full recap

Goal: Explore the proof of scattering for $(i\partial_t - \Delta)u = u^2$

(i) The building blocks: contraction theorems and high-dimensional waves

(ii) Why do simpler approaches fail when studying (2NLS)?

- Slow dispersion in 3d, Insufficient time integrability
- Bilinear range is insufficient in general

(iii) Harmonic analysis: geometry of interacting waves

- Linear wave packets
- Bilinear range is very good in the radial case
- Transversality improves integrability
- Trilinear transversality is better than bilinear transversality

(iv) From harmonic analysis to PDEs

- Strichartz spaces are good for linear transference
- U^2_Δ is great for multilinear transference
- U^2_Δ has bad duality
- V^2_Δ has good duality, but transference is harder

(v) Everything coming together

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(v) Everything coming together

- Resonance analysis
- Check that trilinear interactions are transversal
- Use interpolation to transfer free estimates to V^2_Δ

TL;DR

- Multilinear wave interactions contain extra information beyond the linear theory
- This information comes from transversality and curvature properties of the characteristic surface
- Using multilinear estimates beyond free solutions requires transferring them to larger function spaces
- V^2_Δ is a very suitable transference space in a lot of cases

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Thank you!

Bibliography (1/3)

[GMS09] Pierre Germain, Nader Masmoudi, and Jalal Shatah.
Global Solutions for 3D Quadratic Schrödinger Equations.
International Mathematics Research Notices 2009(3), 414–432 (2009).
doi:10.1093/imrn/rnn135.

[Bej22] Ioan Bejenaru.
The Almost Optimal Multilinear Restriction Estimate for Hypersurfaces with Curvature: The Case of $n-1$ Hypersurfaces in $\mathbb{R}^n$.
International Mathematics Research Notices 2022(20), 16363–16404 (2022).
doi:10.1093/imrn/rnab208.

[BIKT11] Ioan Bejenaru, Alexandru D. Ionescu, Carlos E. Kenig, and Daniel Tataru.
Global Schrödinger Maps in Dimensions $d \geq 2$: Small Data in the Critical Sobolev Spaces.
Annals of Mathematics 173(3), 1443–1506 (2011).
doi:10.4007/annals.2011.173.3.5; arXiv:0807.0265.

[Sha85] Jalal Shatah.
Normal Forms and Quadratic Nonlinear Klein–Gordon Equations.
Communications on Pure and Applied Mathematics 38(5), 685–696 (1985).
doi:10.1002/cpa.3160380516.

[CO17] Myeongju Chae and Sung-Jin Oh.
Small Data Global Existence and Decay for Relativistic Chern–Simons Equations.
Annales Henri Poincaré 18, 2123–2198 (2017).
doi:10.1007/s00023-016-0547-9; arXiv:1512.03039.

[BB26] Ioan Bejenaru and Vitor Borges.
Global Well-Posedness and Scattering for the Massive Dirac–Klein–Gordon System in Two Dimensions.
Communications in Mathematical Physics 407, Article 7 (2026).
doi:10.1007/s00220-025-05499-9; arXiv:2501.03963.
Published online December 6, 2025.

[BC26] Vitor Borges and Tiklung Chan.
Scattering for the Klein–Gordon–Schrödinger System in Three Dimensions with Radial Data.
Preprint (2026).
arXiv:2604.10366.

[Tat01] Daniel Tataru.
On Global Existence and Scattering for the Wave Maps Equation.
American Journal of Mathematics 123(1), 37–77 (2001).
doi:10.1353/ajm.2001.0005.

[Tao01] Terence Tao.
Global Regularity of Wave Maps II. Small Energy in Two Dimensions.
Communications in Mathematical Physics 224(2), 443–544 (2001).
doi:10.1007/PL00005588; arXiv:math/0011173.

[KT17] Joachim Krieger and Daniel Tataru.
Global Well-Posedness for the Yang–Mills Equation in $4+1$ Dimensions. Small Energy.
Annals of Mathematics 185(3), 831–893 (2017).
doi:10.4007/annals.2017.185.3.3; arXiv:1509.00751.

Bibliography (2/3)

[CK93] Demetrios Christodoulou and Sergiu Klainerman.
The Global Nonlinear Stability of the Minkowski Space.
Princeton Mathematical Series, vol. 41. Princeton University Press, 1993.
doi:10.1515/9781400863174.
DOI for later electronic edition.

[LR10] Hans Lindblad and Igor Rodnianski.
The Global Stability of Minkowski Space-Time in Harmonic Gauge.
Annals of Mathematics 171(3), 1401–1477 (2010).
doi:10.4007/annals.2010.171.1401; arXiv:math/0411109.

[GMS12] Pierre Germain, Nader Masmoudi, and Jalal Shatah.
Global Solutions for the Gravity Water Waves Equation in Dimension 3.
Annals of Mathematics 175(2), 691–754 (2012).
doi:10.4007/annals.2012.175.2.6; arXiv:0906.5343.

[IT16] Mihaela Ifrim and Daniel Tataru.
Two Dimensional Water Waves in Holomorphic Coordinates II: Global Solutions.
Bulletin de la Société Mathématique de France 144(2), 369–394 (2016).
doi:10.24033/bsmf.2717; arXiv:1404.7583.

[HTT11] Sebastian Herr, Daniel Tataru, and Nikolay Tzvetkov.
Global Well-Posedness of the Energy-Critical Nonlinear Schrödinger Equation with Small Initial Data in $H^1(\mathbb{T}^3)$.
Duke Mathematical Journal 159(2), 329–349 (2011).
doi:10.1215/00127094-1415889; arXiv:1005.2832.

[IT15] Mihaela Ifrim and Daniel Tataru.
Global Bounds for the Cubic Nonlinear Schrödinger Equation (NLS) in One Space Dimension.
Nonlinearity 28(8), 2661–2675 (2015).
doi:10.1088/0951-7715/28/8/2661; arXiv:1404.7581.

[IT23] Mihaela Ifrim and Daniel Tataru.
Global Solutions for 1D Cubic Defocusing Dispersive Equations: Part I.
Forum of Mathematics, Pi 11, e31 (2023).
doi:10.1017/fmp.2023.30; arXiv:2205.12212.

[IT24] Mihaela Ifrim and Daniel Tataru.
Global Solutions for Cubic Quasilinear Schrödinger Flows in Two and Higher Dimensions.
Preprint (2024), subsequently revised.
arXiv:2404.09970.

[BH17] Ioan Bejenaru and Sebastian Herr.
On Global Well-Posedness and Scattering for the Massive Dirac–Klein–Gordon System.
Journal of the European Mathematical Society 19(8), 2445–2467 (2017).
doi:10.4171/JEMS/721; arXiv:1409.1778.

Bibliography (3/3)

[KST15] Joachim Krieger, Jacob Sterbenz, and Daniel Tataru.
Global Well-Posedness for the Maxwell–Klein–Gordon Equation in $4+1$ Dimensions. Small Energy.
Duke Mathematical Journal 164(6), 973–1040 (2015).
doi:10.1215/00127094-2885982; arXiv:1211.3527.

[RT04] Igor Rodnianski and Terence Tao.
Global Regularity for the Maxwell–Klein–Gordon Equation with Small Critical Sobolev Norm in High Dimensions.
Communications in Mathematical Physics 251(2), 377–426 (2004).
doi:10.1007/s00220-004-1152-1; arXiv:math/0309353.

[IP19] Alexandru D. Ionescu and Benoît Pausader.
On the Global Regularity for a Wave–Klein–Gordon Coupled System.
Acta Mathematica Sinica, English Series 35(6), 933–986 (2019).
doi:10.1007/s10114-019-8413-6; arXiv:1703.02846.

[IP22] Alexandru D. Ionescu and Benoît Pausader.
The Einstein–Klein–Gordon Coupled System: Global Stability of the Minkowski Solution.
Annals of Mathematics Studies, vol. 213. Princeton University Press, 2022.
doi:10.1515/9780691233031; arXiv:1911.10652.

[Wu11] Sijue Wu.
Global Wellposedness of the 3-D Full Water Wave Problem.
Inventiones Mathematicae 184(1), 125–220 (2011).
doi:10.1007/s00222-010-0288-1; arXiv:0910.2473.

[AD15] Thomas Alazard and Jean-Marc Delort.
Global Solutions and Asymptotic Behavior for Two Dimensional Gravity Water Waves.
Annales scientifiques de l'École normale supérieure 48(5), 1149–1238 (2015).
doi:10.24033/asens.2268.

[IP15] Alexandru D. Ionescu and Fabio Pusateri.
Global Solutions for the Gravity Water Waves System in 2D.
Inventiones Mathematicae 199(3), 653–804 (2015).
doi:10.1007/s00222-014-0521-4; arXiv:1303.5357.

[IP18] Alexandru D. Ionescu and Fabio Pusateri.
Global Regularity for 2D Water Waves with Surface Tension.
Memoirs of the American Mathematical Society 256(1227) (2018).
doi:10.1090/memo/1227; arXiv:1408.4428.

[DIPP17] Yu Deng, Alexandru D. Ionescu, Benoît Pausader, and Fabio Pusateri.
Global Solutions of the Gravity-Capillary Water-Wave System in Three Dimensions.
Acta Mathematica 219(2), 213–402 (2017).
doi:10.4310/ACTA.2017.v219.n2.a1; arXiv:1601.05685.